

Recency vs. frequency in induction heads: first run

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In short

- **The setup works.** The Bayes oracle matches brute-force enumeration exactly, the clean-data gate passed, and every model reaches within 0.0004–0.021 nats of the Bayes-optimal loss on ordinary data.
- **When old and recent evidence conflict, the models do not follow the Bayes rule.** As the gap since the old value grows, Bayes shifts its preference gradually. The models hold on, then flip abruptly. Their steepest drop per doubling of the gap is 1.6–3.5× Bayes's for 2-layer models and 1.2–1.8× for 3-layer models, with very small spread across seeds.
- **The learned recency slope tracks the switch rate.** In attention-only models it goes from 0.002 (no switches) to 0.08 and 0.12 (mixed) to 0.57 (switches only), in the same order in every seed.
- **Old copies count more than they should, then saturate.** Adding a second old copy moves the models 2.4–8× more than it moves Bayes. Beyond four copies the effect stops growing.
- **Main caveat:** learnable ALiBi itself makes attention fall off exponentially with distance, so the abrupt flip may come from the positional encoding. A control without ALiBi is the next run.

Verdict: we are on the right path. The models implement something closer to a distance-discounted counter than to the Bayes rule, and the clearest evidence is the shape of the gap curve.

The question

A pattern's continuation appears often but long ago (c. "Potter"), and a conflicting continuation appears rarely but recently (w. "Potver"). Which one does an induction head

...which one appears more likely, the recent copy or the frequent copy? Which one is right depends on how the data was generated. If values really change over time (switch rate h), the recent copy is informative. If copies are only corrupted (typo rate ϵ), the frequent copy is. We compare two candidate rules.

Prop. 1 · discounted counter

A softmax head with a distance penalty $\lambda \cdot d$ weighs each earlier copy by $e^{-\lambda d}$ and adds them up.

pick $c \Leftrightarrow \log(n_c / n_w) > \lambda \cdot (d_c$

Depends on the number of old copies n_c . Distance enters linearly, so preference falls off sharply.

Prop. 2 · Bayes-optimal

Under the switch-and-typo model, the question is whether a switch in the gap G is more likely than n_w identical typos.

pick $w \Leftrightarrow \log(h \cdot G) > \log V - n_w \cdot 1$

Independent of n_c once $n_c \geq 2$. The gap enters as $\log G$, so preference shifts gradually.

Setup

- Task.** Sequences of 256 key–value pairs, 16 keys and 64 values. Each key has a hidden value that is redrawn with probability h per pair step; each observed value is a typo ($z+1$) with probability ϵ . Keys follow a Zipf distribution, so long gaps between sightings of a key occur naturally in training.
- Bayes oracle.** An exact forward filter per key gives the optimal prediction at every position. It is unit-tested against brute-force enumeration over all hidden paths.
- Models.** Pre-LayerNorm transformers, width 128, 4 heads, learnable ALiBi slope per head. Three architectures: 2-layer attention-only (L2), 2-layer with MLPs (L2+MLP), 3-layer attention-only (L3). 20,000 steps on freshly generated data, 3 seeds each.
- Probes.** For a fresh key: n_c copies of c (1, 2, 4, 8), a gap of G pairs (2 to 128), n_w copies of $w = c+1$ (1 to 4), then the query. We record $\log p(c)/p(w)$ for the model and for Bayes, 256 random fills per cell.

Point	h	ϵ	What it tests	Bayes switch-over gap G^* ($n_w = 1, 2, 3$)
A · typos only	0	0.2	never trust w	never
B · switches only	0.03	0	always trust w	0, 0, 0
C · mixed	0.01	0.1	genuine conflict	208, 58, 8

G* in pairs, from the Prop. 2 formula with V = 64. The gate run (h = ε = 0, L2, 2 seeds) reached 100% accuracy on repeat sightings and matched the Bayes floor of 0.259 nats to within 0.0001.

Results

1 · Models reach Bayes on ordinary data

Excess loss over Bayes (nats)	A · typos	B · switches	C · mixed	D · mixed
L2	0.001	0.001	0.016 ± 0.003	0.021 ± 0.002
L2+MLP	0.001	< 0.001	0.004 ± 0.001	0.005 ± 0.000
L3	0.001	< 0.001	0.005 ± 0.000	0.005 ± 0.000

Mean ± sd over 3 seeds, on a fixed evaluation set of 512 sequences. The corners are easy; at the conflict points, MLPs or a third layer cut the remaining gap 3–4×.

2 · In conflicts, the models flip abruptly where Bayes shifts gradually



Log p(c)/p(w) against the gap G, for 3 recent copies of w. Solid lines are models (mean of 3 seeds), dashed lines are Bayes; colors are n_c = 1, 2, 4, 8. Rows 2 and 4 (points C and D) carry the result: the model curves are flat and then fall off a cliff, ending far more confident in w than Bayes.



The same probes with a single recent copy. At point C and a gap of 128, Bayes still keeps the old value (log-odds +0.76) while all models have switched: L2 -2.07, L2+MLP -0.55, L3 -0.63.

Steepest drop per doubling of G	Bayes	L2	L2+MLP	L3
C · h = 0.01, ε = 0.1	0.90	3.14 ± 0.05	2.96 ± 0.13	1.59 ± 0.16
D · h = 0.03, ε = 0.2	1.37	2.25 ± 0.05	2.36 ± 0.06	1.62 ± 0.10

For each (n_c ≥ 2, n_w) curve, the largest fall in log-odds between neighbouring gaps (a factor of 2 apart), averaged over curves; mean ± sd over seeds. The 2-layer models are 1.6–3.5× steeper than Bayes; the 3-layer model is closest.

Switch-over gap G* (n_c = 2)	n_w = 1	n_w = 2	n_w = 3
C · h = 0.01, ε = 0.1			
Bayes	> 128 (formula 208)	52 (formula 58)	6.9 (formula 8.4)

L2	78 ± 25	38 ± 3	20 ± 1
L2+MLP	109 ± 12	41 ± 2	20 ± 2
L3	103 ± 11	36 ± 1	4 ± 4
D · h = 0.03, ε = 0.2			
Bayes	85 (formula 93)	43 (formula 53)	17 (formula 23)
L2	40 ± 2	21 ± 1	12 ± 2
L2+MLP	49 ± 1	31 ± 3	17 ± 1
L3	50 ± 3	24 ± 1	8 ± 0.4

The gap (in pairs) where mean log-odds cross zero, interpolated on a log scale. The measured Bayes boundaries agree with the Prop. 2 formula to within 9–26%, which also checks the theory. At C, the 2-layer models switch too early with one recent copy and too late with three: their boundary moves about 4–5× across $n_w = 1 \rightarrow 3$, against more than 18× for Bayes. At D, all models switch earlier than Bayes with one or two recent copies.

3 · The learned recency slope follows the switch rate

χ of the main induction head (per token)	A · h = 0	C · h = 0.01	D · h = 0.03, ε = 0.2
L2	0.002 ± 0.001	0.078 ± 0.016	0.115 ± 0.005
L3	0.002 ± 0.000	0.056 ± 0.001	0.113 ± 0.005
L2+MLP	0.002 ± 0.000	0.416 ± 0.018	0.466 ± 0.018

The ALiBi slope of the head with the highest induction score (attention from a key to earlier values of the same key). Columns are ordered from no switches to switches only. For attention-only models the slope rises in that order in every seed; for L2 it corresponds to an attention half-life of about 170 pairs at A, 4 at C, 3 at D and 0.6 at B. The MLP models do not follow the order, so one head does not describe them.

4 · Old copies count, then saturate

Change in log p(c)/p(w)	Bayes	L2	L2+MLP	L3
C · h = 0.01, ε = 0.1				
$n_c\ 1 \rightarrow 2$	0.061	0.32 ± 0.11	0.44 ± 0.03	0.49 ± 0.10
$n_c\ 2 \rightarrow 4$	0.008	0.11 ± 0.15	0.20 ± 0.01	0.27 ± 0.07

n_c 4 → 8	0.000	-0.17 ± 0.22	-0.07 ± 0.03	0.03 ± 0.10
D · h = 0.03, ε = 0.2				
n_c 1 → 2	0.075	0.19 ± 0.07	0.18 ± 0.02	0.29 ± 0.05
n_c 2 → 4	0.026	0.06 ± 0.06	0.07 ± 0.02	0.05 ± 0.06
n_c 4 → 8	0.002	-0.10 ± 0.07	-0.05 ± 0.01	-0.05 ± 0.10

Averaged over all (n_w, G) probe cells. For example, for L3 at C with 3 recent copies and a gap of 2, going from 1 to 4 old copies moves the model by 1.55 and Bayes by 0.06. The effect rises and then flattens, as a normalized attention fraction would, rather than growing like log n_c. The small reversal at 8 copies has a large seed spread and is not explained yet.

5 · Point A turned out to be a weak test

With structured typos (w = c+1) and no switches, every model matches Bayes and never flips to w, even with 3 recent copies against 1 old one. A head can learn that a copied w is also evidence for c, which undoes the typo inside the copying step. This sidesteps the counter-versus-Bayes question, so a uniform-typo version is the better frequency test.

Hypotheses written down before the run

(a) 2-layer attention-only models count old copies Adding a second old copy moves them 2.5–5× more than it moves Bayes, but the effect saturates by four copies. That matches Prop. 1 in its normalized form, not the log-count form.	partly
(b) MLPs or a third layer move models toward Bayes on n_c They do get closer to Bayes overall: 3–4× lower excess loss, and the 3-layer gap curve is the closest to Bayes's shape. But their sensitivity to n_c is equal or larger, so the improvement does not come from ignoring stale copies.	not as stated
(c) The learned recency slope rises with the switch rate Monotone in every seed for both attention-only architectures. Not visible in the strongest head of the MLP models.	supported
Not planned: the shape of the gap curve is the clearest discriminator	new

All architectures flip more abruptly than Bayes. The seed spread is at most 0.16, against differences from Bayes of 0.25–2.2. This needs the positional-encoding control below before it can be claimed.

Caveats

ALiBi may be producing the cliff.

Learnable ALiBi adds a penalty proportional to distance to every attention score, which is exactly an exponential distance kernel. The abrupt flip could come from that choice rather than from what the models learned. Models with no positional encoding or learned absolute positions must be checked before this goes in a paper.

- **The conflicts are rare in training.** The models are near Bayes on ordinary data and far from it on the probes, so the deviations live in configurations the training signal rarely reaches. That is still a fact about the learned rule, but more training or conflict-heavy data might shrink it.
- **$\hat{\chi}$ comes from one head.** It describes the attention-only models well and the MLP models poorly.
- **The probe layout is fixed.** Copies are always 4 pairs apart and the query 4 pairs after the last w .
- **One task size, short training.** 16 keys, 64 values, 256 pairs, 20,000 steps, 3 seeds per cell.
- **Prop. 1 not yet fitted directly.** We have not yet checked how often a model's own fitted counter predicts its decisions.

Next steps

1. **Positional-encoding control.** Points C and D with no positional encoding and with learned absolute positions, same 3 architectures $\times$ 3 seeds (36 runs, about 4 hours). If the cliff survives, it is the headline; if it disappears, the result becomes "the positional encoding chooses the rule".
2. **Uniform-typo frequency corner.** Replace point A with uniform typos, where counting really is optimal.
3. **Prop. 1 fidelity.** Fit a discounted counter to the L2 models' own attention and measure how often it predicts their decisions.

4. **Conflict-enriched training.** Check whether more exposure to conflicts moves the models toward Bayes.
5. **Full grid** (5×5 values of h and ϵ) only after steps 1–3.

Reproducing

Code is in `recfreq/` (data generator, exact Bayes oracle, model, training, probes, analysis); 19 unit tests pass, including the oracle against brute-force enumeration. On the server the runs live in `~/recfreq/results/runs/`.

```
python -m pytest -q
python scripts/queue.py configs/E0.json --max-parallel 2    # gate
python scripts/queue.py configs/E1.json --max-parallel 4    # this run
python -m recfreq.analyze results/runs --exp E1 --out figs
```